



Beyond the browser

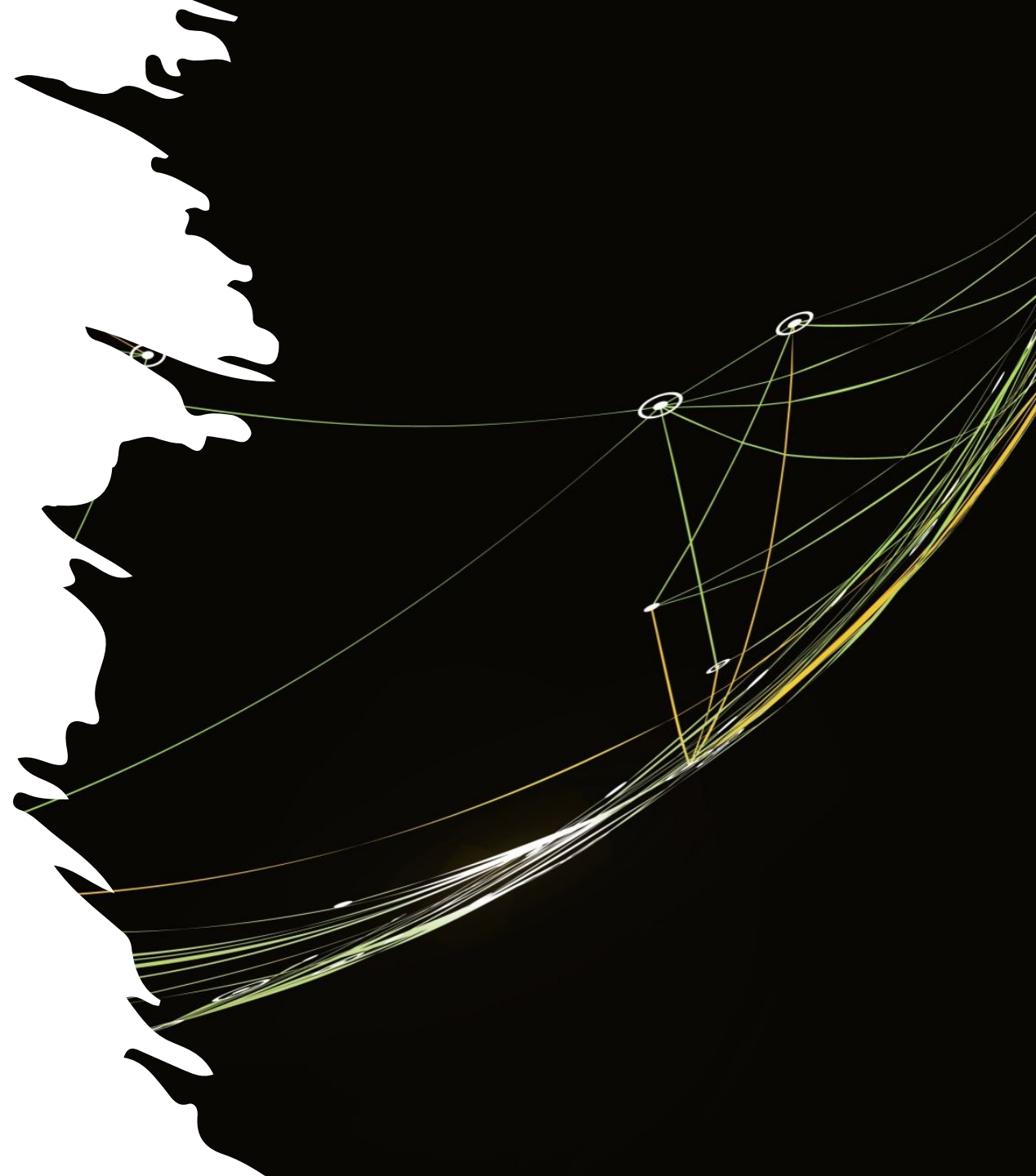
Toward a more declarative future
with WebAssembly

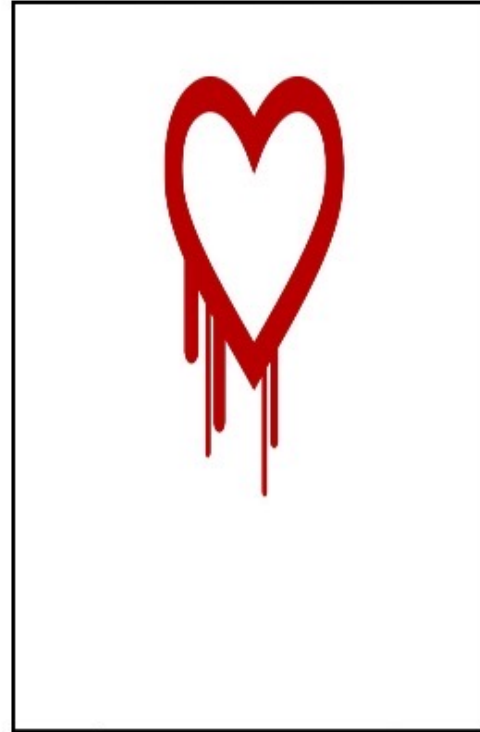


Hi there 🖐️🖐️

- Senior Technical Evangelist @ SUSE
- CNCF Ambassador
- Co-creator of the KCNA certification
- Cloud Native Wasm Day co-chair
- Documentation maintainer for Kubernetes

How we build
software is broken



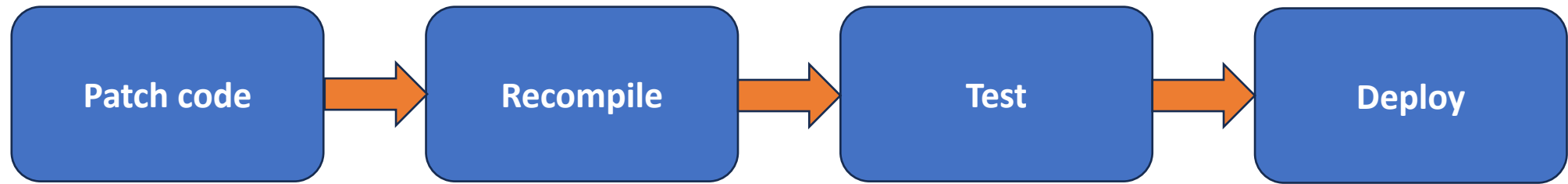


who are you?

imgflip.com

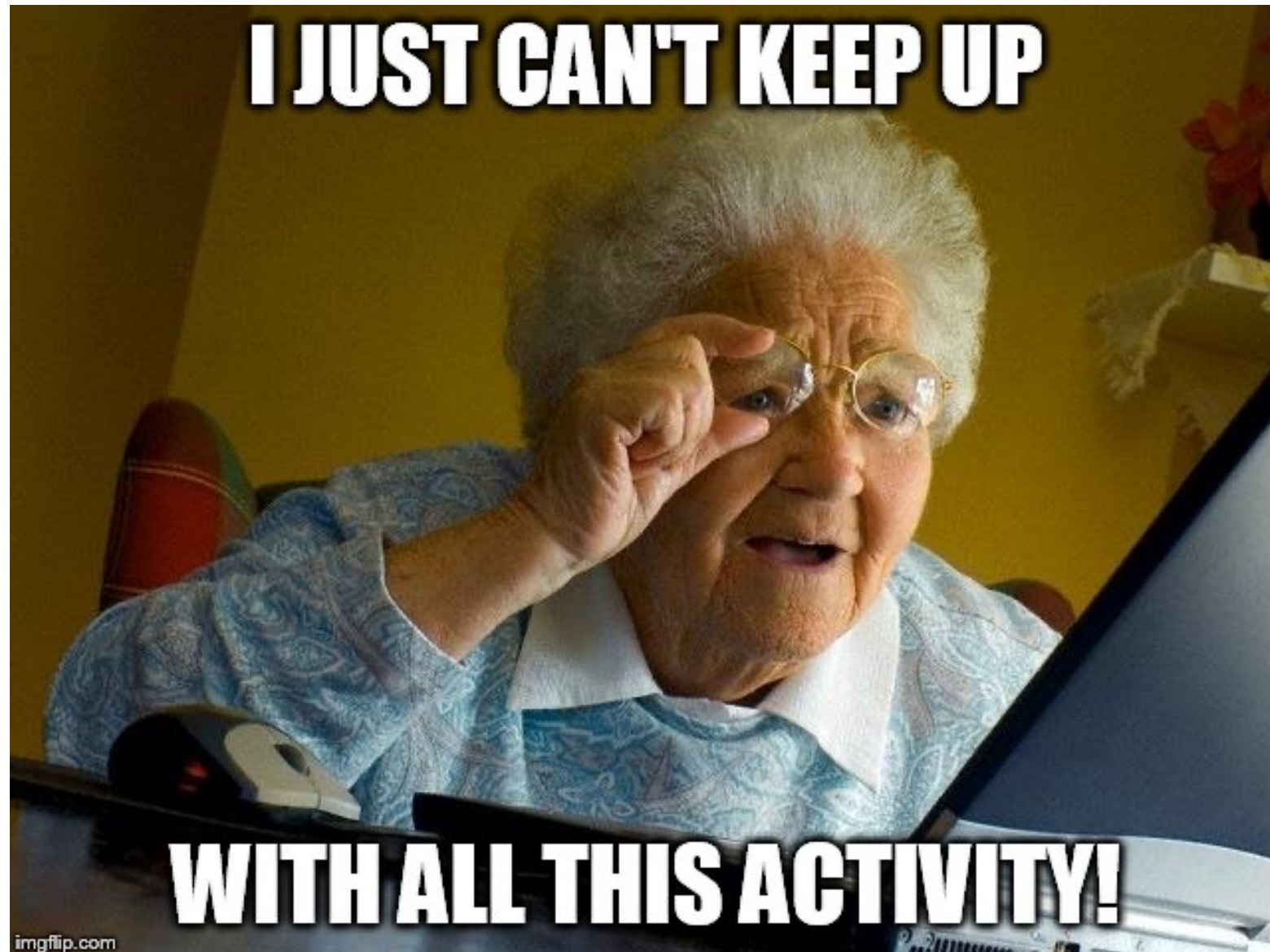
im you but stronger





...



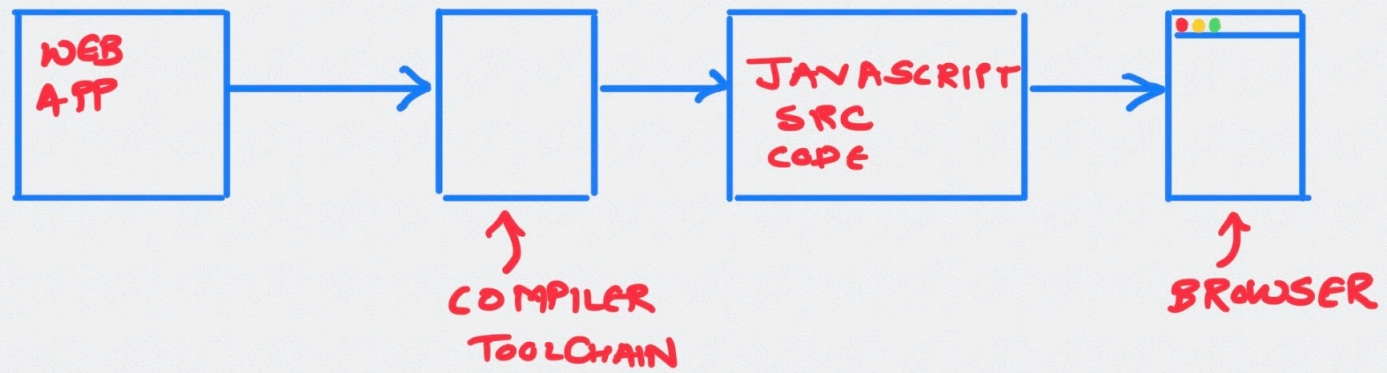


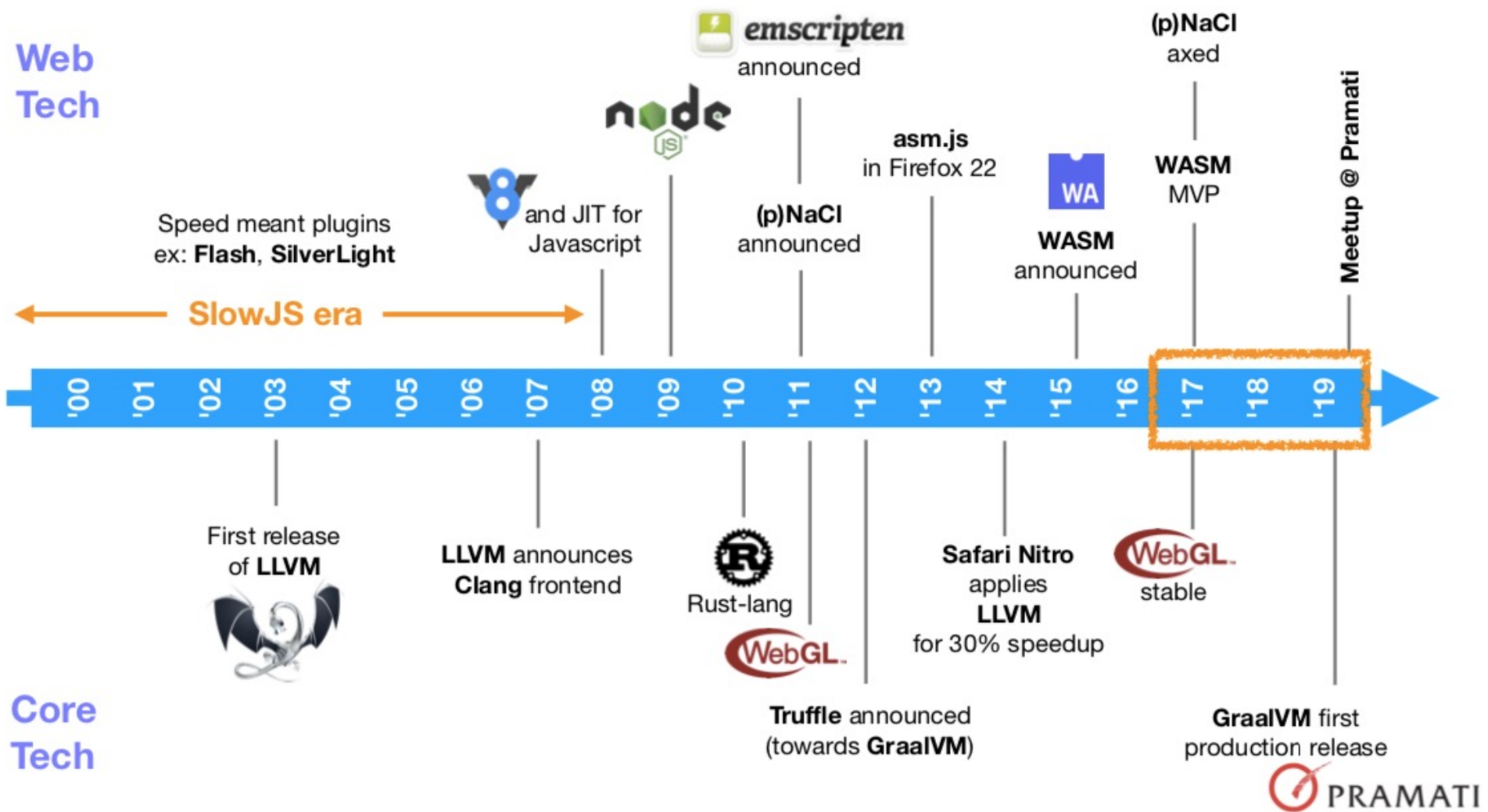
A close-up, dark image of a Mandalorian helmet, likely from the Star Wars franchise. The helmet is metallic and has a distinctive T-shaped visor. The lighting is dim, with some highlights on the helmet's surface.

NOT ONLY
This is [^]the [^]way

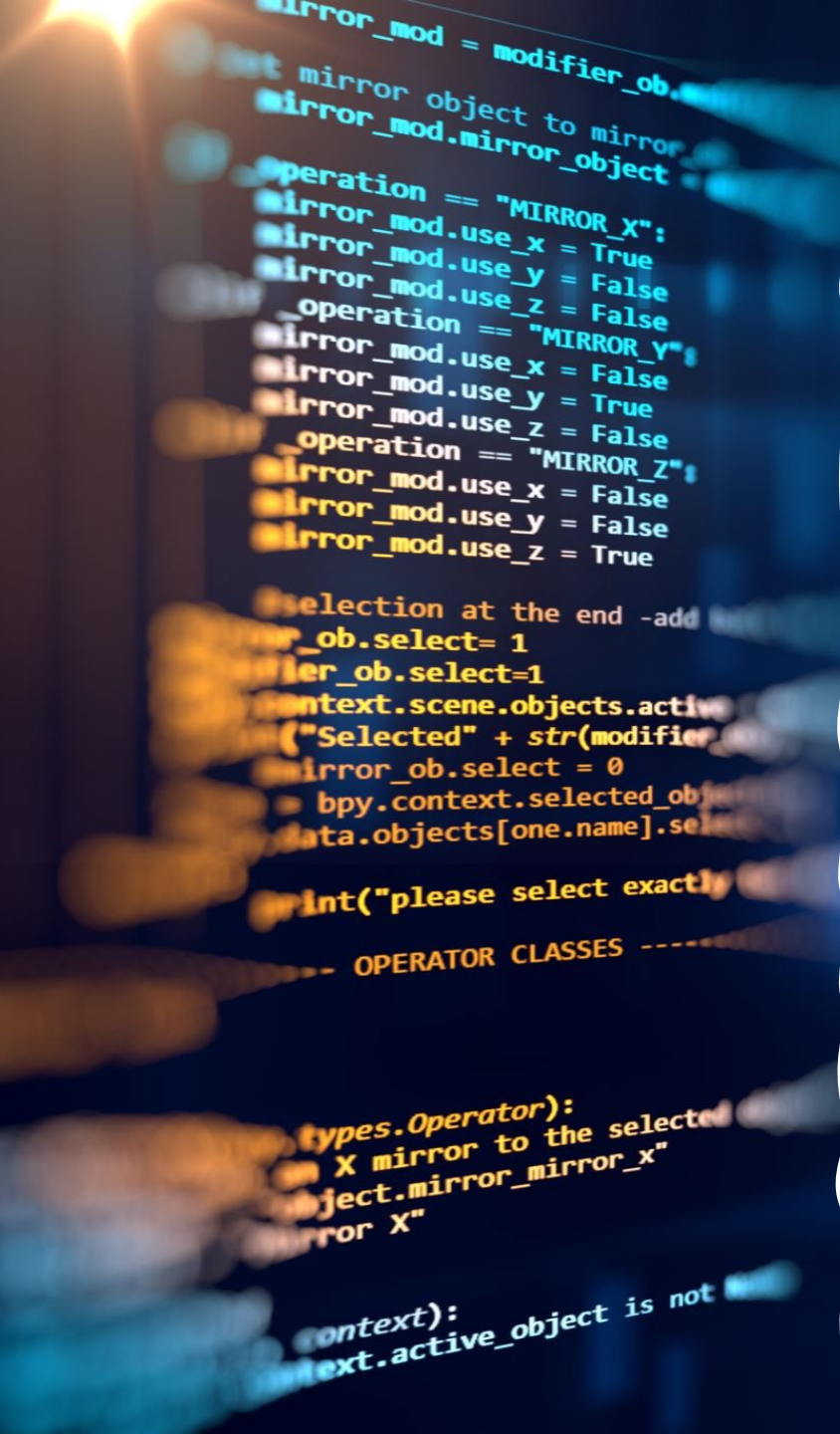
DO YOU HAVE TIME TO TALK ABOUT

WEBASSEMBLY





Source: <http://sriku.org/blog/2019/08/24/talk-the-nuts-and-bolts-of-webassembly/>



“WebAssembly (abbreviated *Wasm*) is a binary instruction format for a stack-based virtual machine. Wasm is designed as a portable compilation target for programming languages, enabling deployment on the web for client and server applications.”

Source: <https://webassembly.org>



What does a Wasm program look like?

```
1 (module)
```

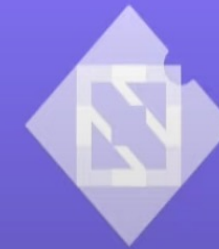
```
                                00 01 02 03 04 05 06 07
00000000 00 61 73 6D 01 00 00 00
```

Wasm Modules

- Compilation target that many languages support
- deny-by-default but really no function to call by default



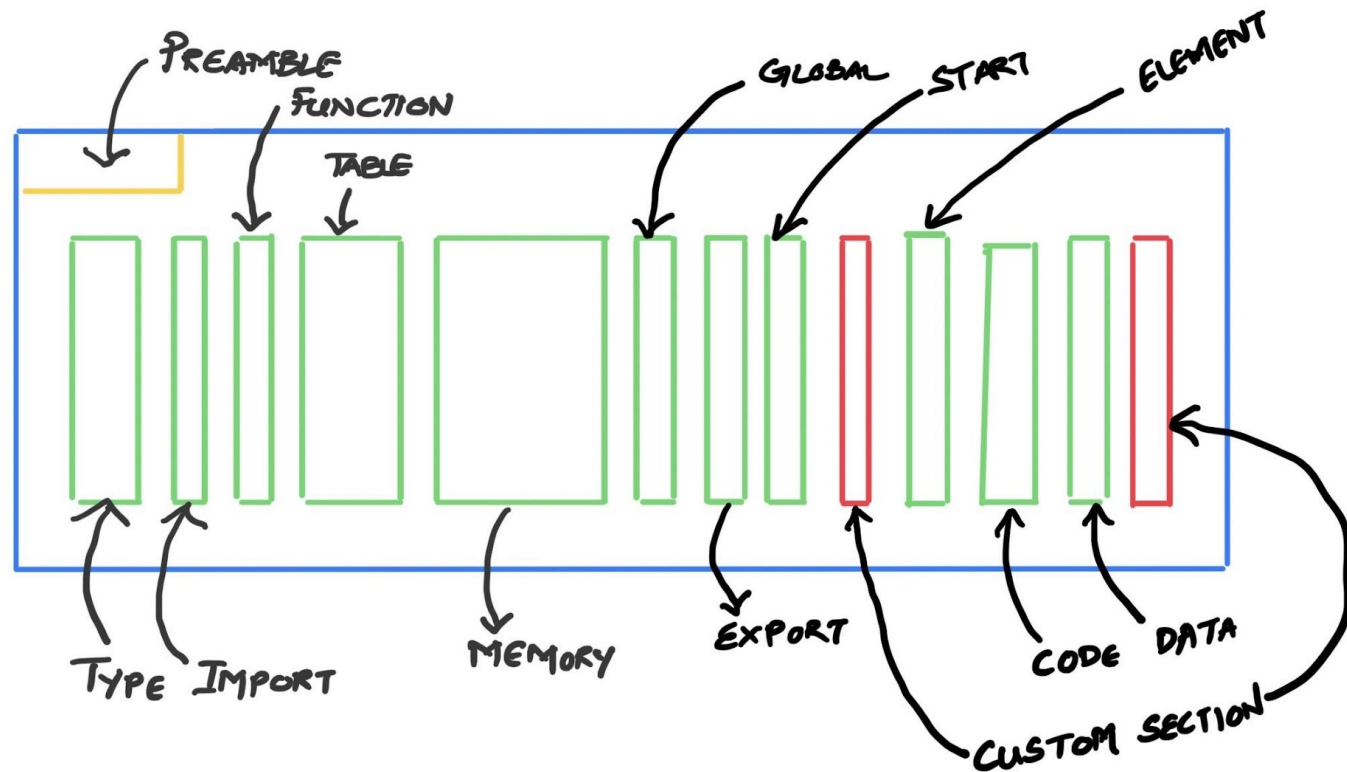
CLOUD NATIVE
Wasm DAY



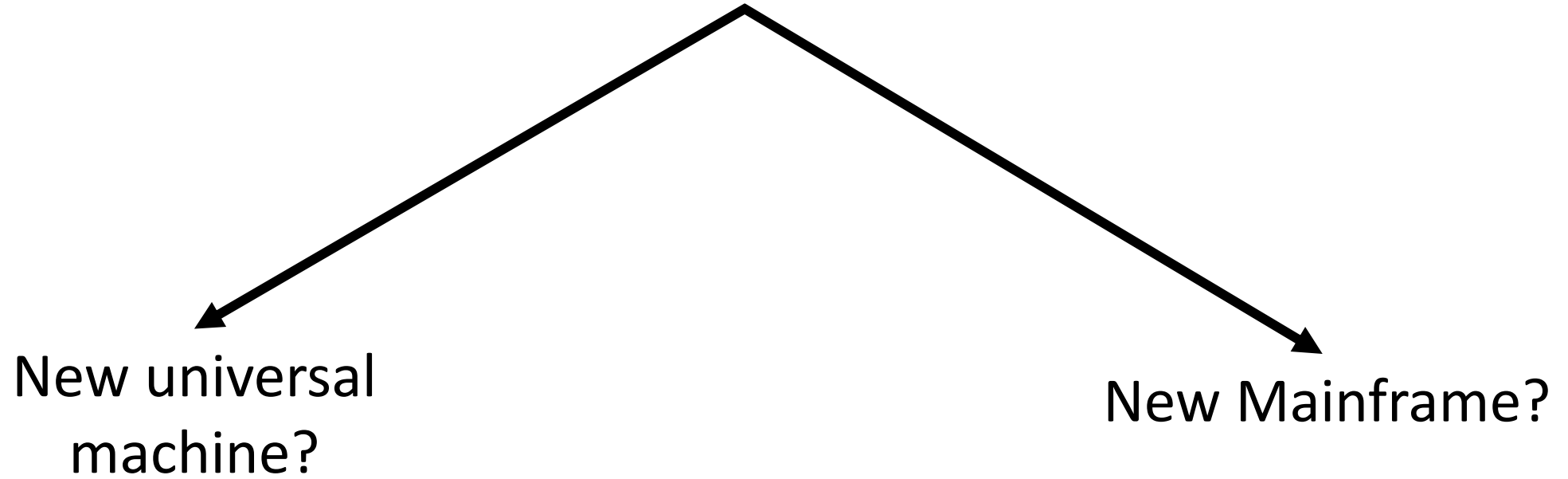
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Source: Evolution of Wasm: Past, Present, Future by Bailey Hayes
https://youtu.be/6_BRLqxiZPU

Anatomy of a Wasm module



WebAssembly



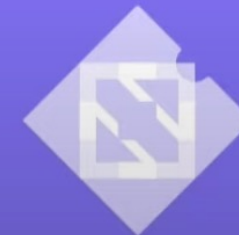
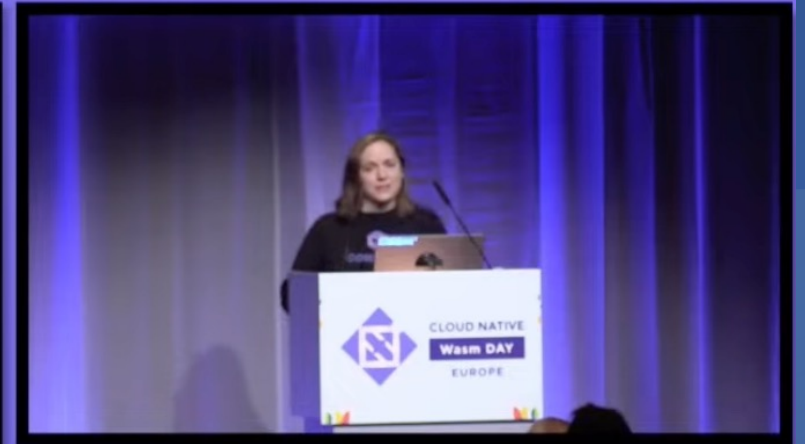
$$21 = \left(\sqrt{\frac{4K(1+K)}{P_{SF}}} P_{\eta} \right)^2 + \kappa^2 \text{ and}$$

Wasm Modules

- Compilation target that many languages support
- deny-by-default but really no function to call by default



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Source: Evolution of Wasm: Past, Present, Future by Bailey Hayes
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WEBASSEMBLY

[Overview](#)[Getting Started](#)[Specs](#)[Future features](#)[Community](#)[FAQ](#)

WebAssembly 1.0 has shipped in 4 major browser engines.

[Learn more](#)

Specifications

- [Core specification](#): defines the semantics of WebAssembly modules independent from a concrete embedding. The WebAssembly core is specified in a single document.
- Embedding interfaces:
 - [JavaScript API](#): defines JavaScript classes and objects for accessing WebAssembly from within JavaScript, including methods for validation, compilation, instantiation, and classes for representing and manipulating imports and exports as JavaScript objects.
 - [Web API](#): defines extensions to the JavaScript API made available specifically in web browsers, in particular, an interface for streaming compilation and instantiation from origin-bound `Response` types.
 - [WASI API](#): defines a modular system interface to run WebAssembly outside the web, providing access to things like files, network connections, clocks, and random numbers.
- [Tool conventions](#): repository describing non-standard conventions useful for coordinating interoperability between tools working with WebAssembly. This includes conventions for linking schemes, debugging information, language ABIs and more.
- [Original design documents](#): documents describing the design, goals and high-level overview of WebAssembly. Some of these documents are outdated by now.

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BYTECODE ALLIANCE

About the Bytecode Alliance

The Bytecode Alliance is a nonprofit organization dedicated to creating secure new software foundations, building on standards such as [WebAssembly](#) and [WebAssembly System Interface \(WASI\)](#).

CNCF Cloud Native Interactive Landscape

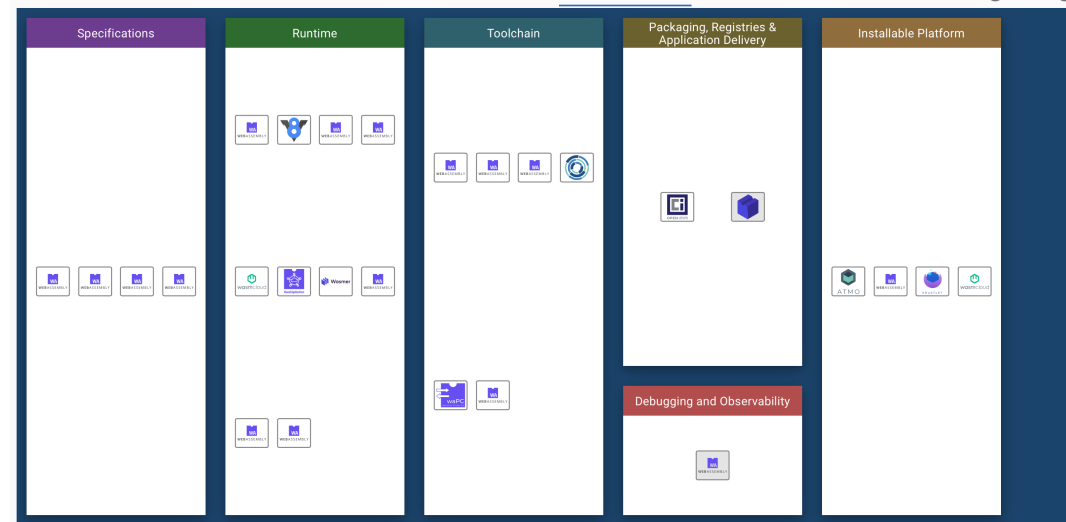


The cloud native landscape (png, pdf), serverless landscape (png, pdf), and member landscape (png, pdf) are dynamically generated below. Please open a pull request to correct any issues. Greyed logos are not open source. Last Updated: 2023-05-29T13:35:28.

You are viewing 27 cards with a total of 140,375 stars, market cap of \$21.5T and funding of \$53.4B.

[Landscape](#)[Card Mode](#)[Members](#)[Serverless](#)[Wasm](#)

120%





The Component Model



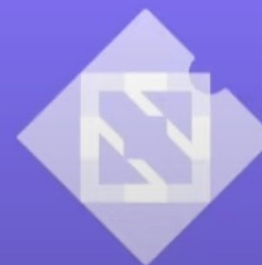


What is a
component?





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Wasm DAY



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Wasm DAY
EUROPE

Hello

- Describes imports & exports in WIT
- MAY represent the execution environment of a component.

Hello

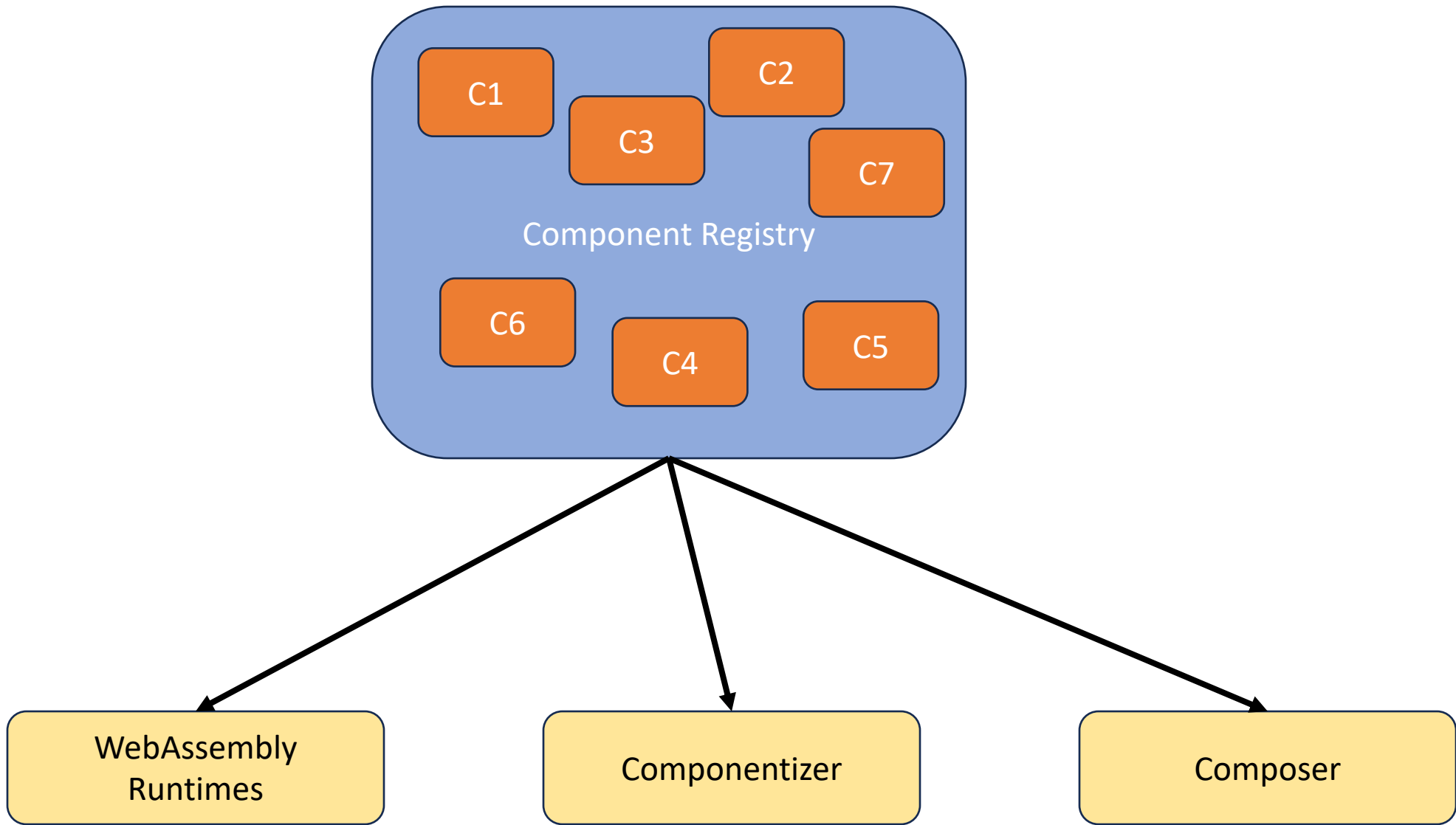
```
world my-world {  
  import host: interface {  
    log: func(param: string)  
  }  
  
  export run: func()  
}
```



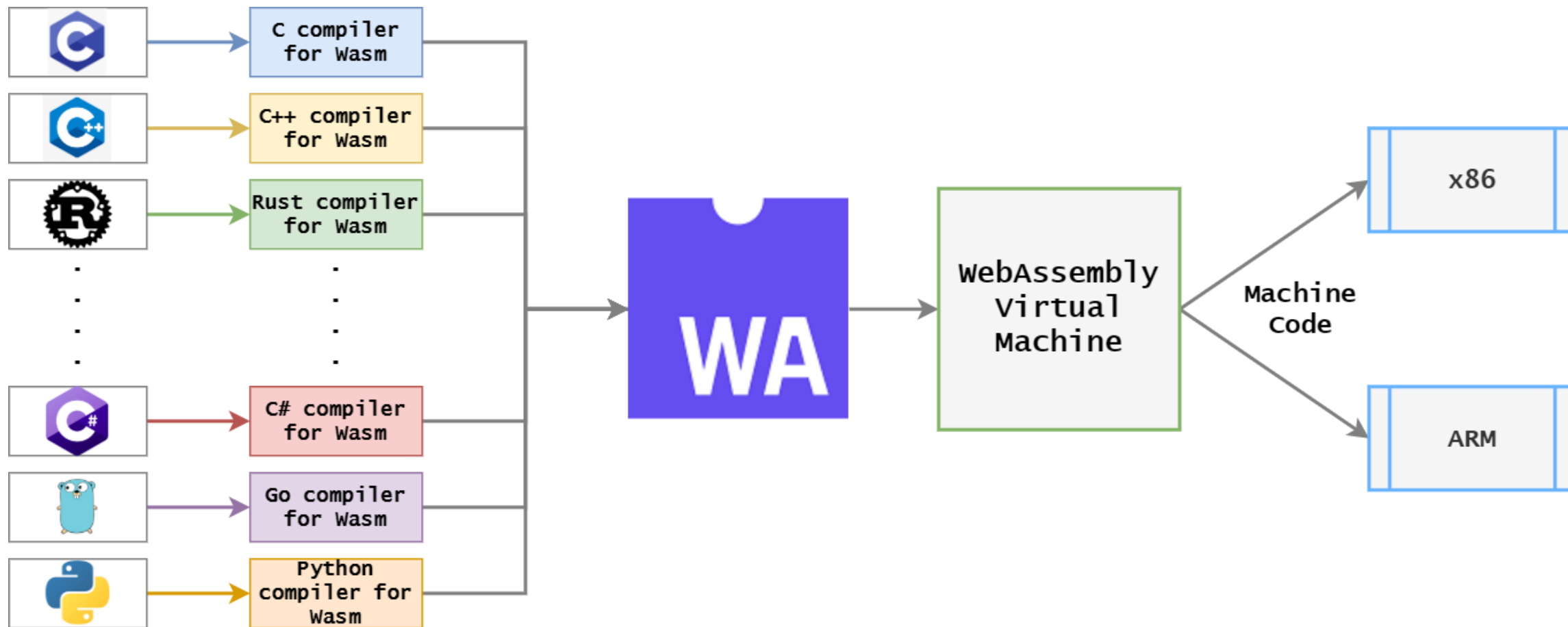
```
(type $my-world (component  
  (import "host" (instance  
    (export "log" (func (param "param" string)))  
  ))  
  (export "run" (func))  
))
```

Component Registry

- Federated namespace of Wasm components
- Implements package transparency to enhance security
 - Cryptographic verifiable log of anything that has ever happened to a component



Benefits?



Demo time!

Resources

- Bytecode Alliance: <https://bytecodealliance.org>
- The Component Model:

<https://github.com/WebAssembly/component-model/tree/main>

